SUPPOSITORIES

Anusol Plus SUPPOSITORIES
WITH ANESTHETIC
FAST, EFFECTIVE HEMORRHOID RELIEF
• Helps Reduce Swelling
• Relieves Itch, Burning & Irritation
• Extra Pain Relief
24
#1 DOCTOR RECOMMENDED BRAND

FAST RELIEF
SOULAGEMENT RAPIDE
Dulcolax
Bisacodyl suppositories / suppositoires
10 mg
Mfr. Std. / Norme-fabricant
#1
LAXATIVE / LAXATIF
Effective and dependable relief
Soulagement fiable et efficace
SOULAGEMENT EN
15-60 MINUTES
SUPPOSITORIES
SUPPOSITOIRES

Topics

- Definition
- Types of suppositories
- Types of bases
- Advantages and disadvantages
- Method of preparations
- Evaluation of suppositories

- Suppositories were derived from the Latin word “Supponere” which means "to place under”
- Suppositories are solid dosage forms intended for insertion into body cavities or orifices (Rectum, Vagina, Nasal, Ear & Urethra) (other than the mouth) where they melt (at body temperature) or dissolve & exert a localized or systemic effect.
- They also come under semisolid dosage form because it is prepared by melting all ingredients like ointments and creams.

Advantages

- They can exert both local and systemic action
- It is used to promote evacuation of the bowel (by using rectal suppositories)
- It avoids any gastrointestinal irritation.
- They can be used in unconscious patients who cannot swallow the medicaments
- Pediatrics and geriatric patients can easily use who cannot swallow oral medicaments
- It is useful for post-operative patients.
- Suitable for people suffering from severe nausea and vomiting.
- Suitable for drugs that can degrade in gastrointestinal irritation

Disadvantages

- Suppositories have problems with patient acceptability
- Manufacturing of suppositories is more difficult and more costly than some dosage form
- Not suitable for drugs that irritate the mucous membrane
- Suppositories have more complicated storage conditions than some dosage forms.
- Suppositories (rectal) are not suitable for patients suffering from diarrhea

Types of Suppositories

- Rectal Suppositories
- Vaginal suppositories (Pessaries)
- Urethral Suppositories (Bougies)
- Nasal Suppositories (Nasal Bougies)
- Ear cones (Aurinaria)

Rectal Suppositories

- These are meant for insertion into the **rectum** for producing local or systemic action.
- These suppositories are usually **32 mm (1.5 inches)** long, cylindrical, bullet or torpedo-in shape, meant for **adults** usually weighing **2 g**, and for **children** are **about half in weight and size.**
- They are used for analgesic, antispasmodic, sedatives, antiseptic, local anesthetic, evacuation of the bowel, or for lubricating the membrane.

Vaginal suppositories

- The vaginal suppositories are also known as **Pessaries.**
- They are meant for insertion into the **vaginal cavities**.
- They weigh about **3-5 gm** and are molded in **globular or oviform shapes**, or on a tablet pressed into **conical shapes**.
- The vaginal suppositories are larger than the rectal.
- They are used for their local action in the vagina as antiseptic
- Medical termination of pregnancy.

Urethral suppositories

- These are also called **Bougies** and are of the pencil shape
- The urethral suppositories are meant for insertion into the male and female urethra.
- The urethra suppositories intended for **males weigh 4 gm** each and are 100-150 mm long, 3-6 mm in diameter and those for **females are 2 gm** each and 60 - 75 mm long.

Nasal suppositories

- The nasal suppositories are also called **nasal bougies (buginaria).**
- The nasal suppositories are meant for introduction into the nasal cavity.
- They are usually prepared with a glycerol-gelatin base.
- They have a similar shape as that of the urethral bougies.
- They weigh about **1 gm** and have a **9-10 mm length**.

Ear Cones

- Ear cones are used for insertion into the ear.
- They are also known as **Aurinaria**.
- They are used rarely.
- For the preparation of ear cones generally Theobroma oil is used as a base

Types of Suppositories Bases

Suppositories bases play an important role in maintaining the shape, solidity, and size of suppositories as well as help in the release of drugs in the body

IDEAL PROPERTIES

- It should melt at body temperature or dissolve in body fluids.
- Good in appearance
- Non-toxic and non-irritant
- Compatible with any medicaments
- Should retain shape while storage and handling.

Classification

- **Fatty Base/Oleaginous Base/Oily Base**
    - Theobroma Oil (Cocoa Butter)
    - Emulsified Theobroma Oil
    - Hydrogenated Base
- **Water Soluble or Water Miscible Base**
    - Glycero-gelatin
    - Polyethylene Glycol (PEG base) (Carbowaxes, Macrogols, Postonals)
- **Emulsifying Base**
    - Massa Esterinum
    - Witepsol
    - Massupol

FATTY BASE/OLEAGINOUS BASE/OILY BASE

Theobroma Oil (Cocoa Butter)

- Yellowish white in color
- Obtained from crushed and roasted seeds of Theobroma cocoa
- Butter-like consistency having a melting point of 30-35 ^{o}C.
- Available in 4 forms
    - **α – form:** M.P. 24 ^{o}C
    - **β – form:** M.P. 34-36 ^{o}C
    - **β^1 – form:** M.P. 28-31 ^{o}C
    - **γ – form:** M.P. 18 ^{o}C
- It is a mixture of glyceryl ester of stearic acid, oleic acid, palmitic acid, and other fatty acids.

Disadvantages:

- Become rancid (oxidation of fats) (unpleasant taste and smell) in warm weather
- Costly
- Immiscible with body fluids
- Stick to the side of the molds when solidified.
- Poor water-absorbing ability

FATTY BASE/OLEAGINOUS BASE/OILY BASE

Emulsified Theobroma Oil

- This base can be used when large quantities of the aqueous solution are to be added.
- Many substances are used to form Emulsified Theobroma Oil
    - 2-3% Cetyl Alcohol
    - 4% Glyceryl Monostearate
    - 10% Lanette wax
    - 4% Beeswax

FATTY BASE/OLEAGINOUS BASE/OILY BASE

Hydrogenated Oil

- This base is used as a substitute for Theobroma oil
- This base is obtained by hydrogenation of Vegetable oils like coconut oil, Palm kernel oil, hydrogenated edible oil, cotton seed oil
- They are also known as synthetic fat bases.
- Produce colorless, odorless suppositories
- Water absorbing capacity is good

Disadvantages:

- Become brittle if cooled rapidly
- Melted fats are less viscous and more fluid than Theobroma oil, the sedimentation rate of the drug is high during the preparation of suppositories.

WATER SOLUBLE OR WATER MISCIBLE BASE

Glycero Gelatin Base (glycerin + water + gelatin)

- This base is a mixture of water and glycerin which is made into stiff jelly by adding gelatin
- Dissolve in body fluid and release medicament
- Proportion of gelatin may vary according to the intended use of preparation
- **Glycero Gelatin Mass BP:** 14% gelatin and 70% glycerin
- **Glycero Gelatin Mass USP:** 20% gelatin and 70% glycerin
- **Pharmagel A / Type A:** acidic in nature and used for acidic drugs
- **Pharmagel B / Type B:** basic in nature and used for basic drugs
- Suppositories of this base are difficult to prepare, handle, and store
- Incompatible with some drugs.

WATER SOLUBLE OR WATER MISCIBLE BASE

PEG Bases

- Different mixtures of polyethylene glycols are marketed under the trade name of Carbowaxes, Macrogols, and Postonals.
- Most of the drugs commonly administered in suppository form using these bases are compatible.
- Chemically stable, non-irritant, and does not allow bacterial growth.

EMULSIFYING BASE

- Massa Esterinum, Witepsol, and Massupol are the trade names under which emulsifying bases are marketed.
- **Massa Esterinum:** it is a mixture of the mono, di, and triglycerides of the fatty acid
- **Witepsol:** it consists of hydrogenated triglycerides of lauric acid with monoglycerides
- **Massupol:** consists of glyceryl ester lauric acid and a small quantity of glyceryl monostearate.

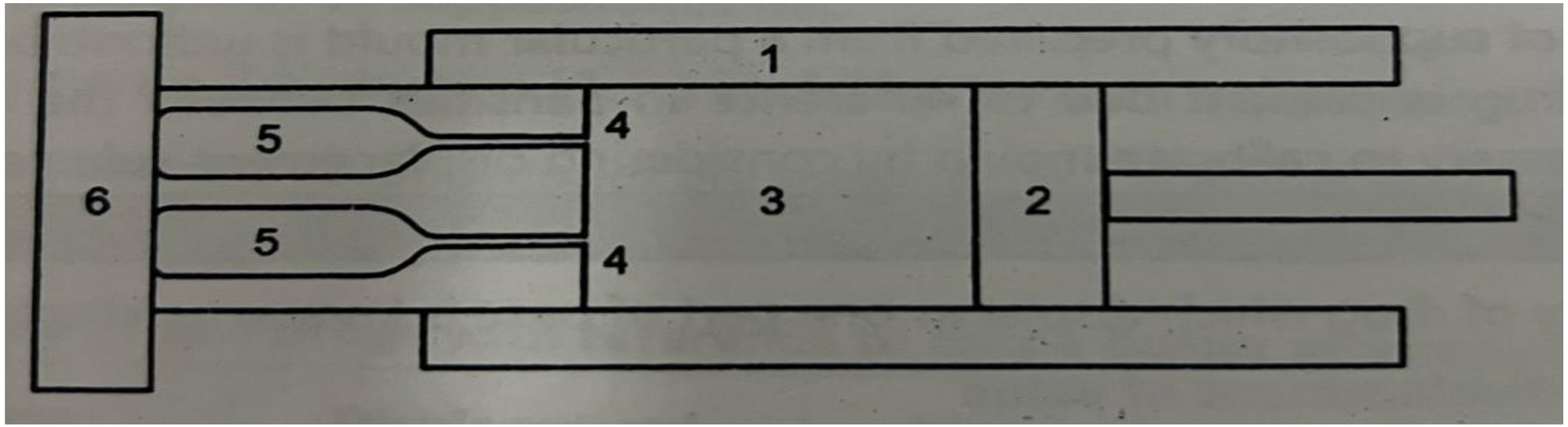
METHOD OF PREPARATIONS

- Hand rolling method
- Compression method
- Molding method / Hot process method

HAND ROLLING METHOD

- It is the simplest and oldest method of suppository preparation and may be used when only a few suppositories are to be prepared in a cocoa butter base.
- It has the advantage of avoiding the necessity of heating the cocoa butter.
- By triturating grated cocoa butter and active ingredients in a mortar a **thick mass** is prepared.
- The mass is formed into a ball in the palm of the hands, then rolled into a uniform cylinder with a large spatula or small flat board.
- The cylinder is then cut into the appropriate number of pieces which are rolled on one end to produce a conical shape.

COMPRESSION METHOD



Cold Compression Machine: 1. Cylinder, 2. Piston, 3. Prepared mass to be compressed, 4. Narrow opening, 5. Moulds, 6. Retaining stop plate

- In this method, the drug is first mixed with a suppository base and then the remaining quantity of base is added and blended thoroughly to form a **mass**.
- This mass is placed into the cylinder. The piston is pushed against suppository mass contained in a cylinder so that mass is forced into the molds through a narrow opening.

MOLDING METHOD / HOT PROCESS METHOD

- It is a hot process in which the suppository base is first melted.
- Incorporating any required medicaments which are either emulsified or suspended in it.
- This molten mass is then transferred to a pre-lubricated suppository mold
- Allowing the melt to cool and congeal (become semisolid upon cooling) into suppositories at a suitable temperature.
- Removing prepared suppositories from the molds.
- It is a widely used method for the preparation of suppositories in both laboratory and industry.

EVALUATION OF SUPPOSITORIES

1. Appearance
2. Uniformity of weight (weight variation) (weight of individual suppositories)
3. Uniformity of content (weight of the drug in each suppository)
4. Melting range test
5. Softening time/liquefaction test
6. Breakage test/breaking time test
7. Dissolution test
8. Stability studies